

MSview

Mass Spectrum Viewer — Tester Quick Start Guide

Beta / Internal Testing Release · March 2026

■ **CONFIDENTIAL — DO NOT SHARE**

This document and the MSview software are for internal testing only. Please do not forward, distribute or share with anyone outside your testing group. A wider release will follow once testing is complete.

1 Installation

Requirements: macOS 10.14+ or Windows 10+ · Python 3.9 or later · Internet connection for first run only

macOS	Windows
1. Unzip msview-v1.zip 2. Open Terminal 3. Type bash followed by a space 4. Drag run_mac.sh into the Terminal window 5. Press Return 6. Wait for dependencies to install (first run only — ~1 min) 7. MSview will open automatically	1. Unzip msview-v1.zip 2. Double-click run_windows.bat 3. If Windows Defender appears, click <i>More info</i> → <i>Run anyway</i> 4. Wait for dependencies to install (first run only — ~1 min) 5. MSview will open automatically

■ On macOS, typing 'bash' before dragging the script bypasses any permission restrictions. Do not double-click the .sh file directly.

2 Loading Spectrum Data

Click the **File** tab in the left panel, then click **Add spectrum...** to open a file.

Format	Extension	Notes
Bruker DataAnalysis XML	.xml	Auto-detected, centroid mode set automatically
mzML	.mzml / .xml	Profile or centroid — auto-detected
mzXML	.mzxml	Auto-detected
Mascot Generic Format	.mgf	Centroid — auto-detected
Plain text / CSV	.xy .txt .csv .ascii	Two-column: m/z intensity

- The app **automatically switches** between Line (profile) and Stick (centroid) display when loading a file.
- You can **overlay multiple spectra** — click Add spectrum... again and each file gets its own colour.
- Double-click a spectrum in the list to **show/hide** it. Select it and click **Change colour** to recolour.

3 Navigating the Interface

Tab	What it does
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File	Load, overlay and manage spectra. Set ion mode (positive/negative).
Display	Switch between Line (profile) and Stick (centroid). Normalise each spectrum to 100%.
Ruler	Activate the $\Delta m/z$ ruler: click two peaks to measure mass difference. Common neutral losses are identified automatically.
Pattern	Enter a molecular formula to calculate and overlay the theoretical isotope pattern. Adjust charge state, peak width and scale.
Annotations	Click any peak to label it, or add labels manually. Auto-label the top 5 or top 10 peaks. Add free-floating text labels anywhere on the plot.

Plot controls:

- **Scroll wheel** — zoom X axis
- **Right-click drag** — zoom to a region
- **Left-click drag** — pan
- **Scroll zoom buttons** (toolbar) — toggle between X-only, Y-only, or both axes
- **Reset view** — fit all data back into view

4 Exporting Results

- **Copy to clipboard** — copies the current view as a PNG, ready to paste into Word, PowerPoint, etc.
- **Save PNG...** — saves the current plot view as a high-resolution PNG file.
- Annotations are included in both exports.

■ *Overlaid spectra, isotope patterns, ruler measurements and free-text labels are all captured in exports.*

5 What We'd Like You to Test

Please try each of the following and note any issues, unexpected behaviour or suggestions:

File loading	Try opening your own .xml, .mzml, .mgf and plain-text spectrum files. Does the mode switch correctly?
Multi-spectrum overlay	Load two or more spectra. Toggle visibility, change colours, check the plot looks correct.
Ruler tool	Activate the $\Delta m/z$ ruler, click two peaks. Does the mass difference and neutral loss hint display correctly?
Isotope pattern	Enter a formula (e.g. C ₁₂ H ₂₂ O ₁₁) in the Pattern tab and overlay it. Scale it to match a peak.
Annotations	Click peaks to label them. Try auto-labelling Top 5/Top 10. Add a free-text label at an arbitrary position.

Zoom & pan	Use scroll, right-drag and the toolbar buttons. Does navigation feel responsive?
Export	Copy to clipboard and paste into another app. Save PNG and check the output.
General	Any crashes, confusing UI, missing features, or anything that didn't work as expected.

6 Reporting Issues

When reporting a bug or suggestion, please include:

- What you were doing when the issue occurred
- What you expected to happen
- What actually happened (include any error message shown)
- Your operating system and Python version
- The file you were loading, if relevant

Thank you for taking the time to test MSview — your feedback is invaluable!